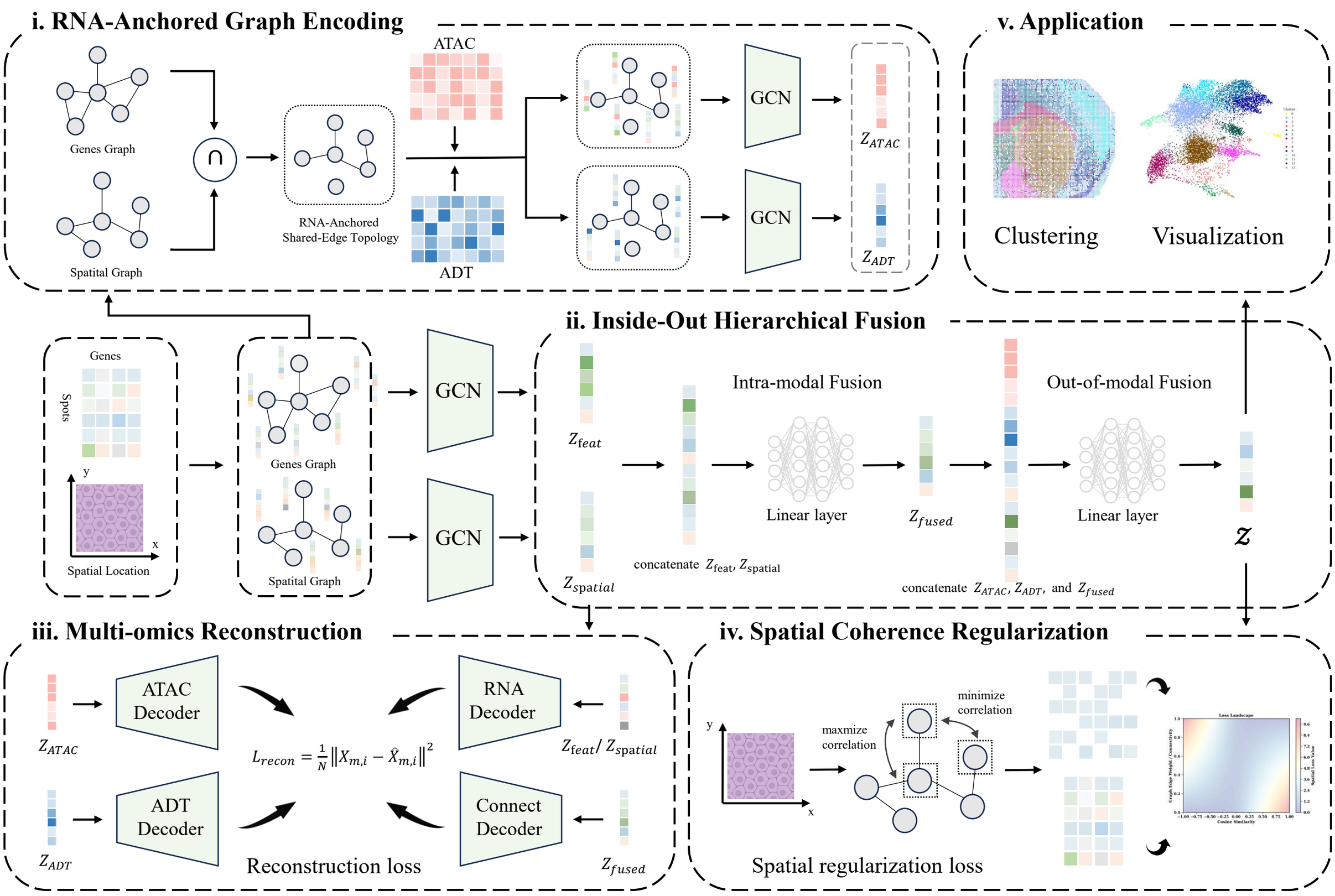
i. RNA-Anchored Graph Encoding
Genes Graph
Spatital Graph
RNA-Anchored
Shared-Edge Topology
ATAC
ADT
GCN
GCN
Z_{ATAC}
Z_{ADT}
v. Application
Clustering
Visualization
ii. Inside-Out Hierarchical Fusion
Genes
Spots
y
x
Spatial Location
Genes Graph
Spatital Graph
GCN
GCN
Z_{feat}
$Z_{spatial}$
Intra-modal Fusion
concatenate Z_{feat}, $Z_{spatial}$
Linear layer
Z_{fused}
Out-of-modal Fusion
concatenate Z_{ATAC}, Z_{ADT}, and Z_{fused}
Linear layer
Z
iii. Multi-omics Reconstruction
Z_{ATAC}
ATAC Decoder
Z_{ADT}
ADT Decoder
$L_{recon} = \frac{1}{N}\|X_{m,i} - \hat{X}_{m,i}\|^2$
RNA Decoder
Connect Decoder
Z_{feat}/ $Z_{spatial}$
Z_{fused}
Reconstruction loss
iv. Spatial Coherence Regularization
y
x
maxmize correlation
minimize correlation
Spatial regularization loss